



Lab Report

Lab Report No : 05							
Lab Report Name : Concise Comparison of Supervised Machine Learning Models for Stroke Prediction							
Course Title : Artificial Intelligence Sessional							
Course Code : CSE 422							
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ID : 0822220105101024							
Level :	4	Term :	2	Section :	A	Group :	G1
Date of Submission :	05/07/2026	Semester :	Spring	Year :	2026		

Marking Rubric

Problem Understanding & Report Clarity (3)	Implementation (4)	Results, Analysis & Presentation (3)	Total (10)

Key Learnings:

- Learned to clean real-world healthcare datasets by handling missing values, converting categorical features into numerical values, and applying feature scaling, which is especially important for logistic regression and SVM.
- Understood how imbalanced classes affect model performance and learned that oversampling improves learning by giving the minority class adequate representation during training.
- Gained experience splitting data into training and testing sets, then implementing four supervised machine learning algorithms to build predictive models.
- Learned to evaluate models using accuracy, precision, recall, and F1-score and understood how a confusion matrix helps assess the correctness of predictions.
- Learned how to objectively compare multiple models using both numerical tables and visualizations to identify the best-performing approach.

Sample Output:

```
Supervised-ML.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
RAM Disk
22s
STEP 1: DATASET OVERVIEW
=====
Total rows : 5110
Total columns : 12

Columns:
['id', 'gender', 'age', 'hypertension', 'heart_disease', 'ever_married', 'work_type', 'Residence_type', 'avg_glucose_level', 'bmi', 'smoking_status', 'stroke']

Target (stroke) distribution:
stroke
0    4861
1     249
Name: count, dtype: int64

Missing values:
id            0
gender        0
age           0
hypertension  0
heart_disease 0
ever_married  0
work_type     0
Residence_type 0
avg_glucose_level 0
bmi           261
smoking_status 0
stroke        0
dtype: int64
=====
```

Figure No: 01

Supervised-ML.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

RAM Disk

```
=====
STEP 2: DATA PREPROCESSING
=====
...
✓ Dropped 'id' column
✓ Removed 1 row with gender='Other'
✓ Filled 201 missing BMI values with median (28.1)
✓ Label encoded columns: ['gender', 'ever_married', 'work_type', 'Residence_type', 'smoking_status']

=====
STEP 3: HANDLE CLASS IMBALANCE
=====
Before balancing: {0: 4860, 1: 249}
After balancing: {0: 4860, 1: 4860}

=====
STEP 4: SPLIT DATA
=====
Training samples : 7776
Testing samples : 1944

=====
STEP 5: TRAINING 4 MODELS
=====

Logistic Regression
```

Variables Terminal

11:23 PM Python

Figure No: 02

Supervised-ML.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

RAM Disk

```
=====
STEP 2: DATA PREPROCESSING
=====
...
✓ Dropped 'id' column
✓ Removed 1 row with gender='Other'
✓ Filled 201 missing BMI values with median (28.1)
✓ Label encoded columns: ['gender', 'ever_married', 'work_type', 'Residence_type', 'smoking_status']

=====
STEP 3: HANDLE CLASS IMBALANCE
=====
Before balancing: {0: 4860, 1: 249}
After balancing: {0: 4860, 1: 4860}

=====
STEP 4: SPLIT DATA
=====
Training samples : 7776
Testing samples : 1944

=====
STEP 5: TRAINING 4 MODELS
=====

Logistic Regression
```

Variables Terminal

11:23 PM Python

Figure No: 03

Supervised-ML.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

RAM Disk

```
Random Forest
=====
              precision    recall  f1-score   support

No Stroke      1.00        0.98        0.99         972
Stroke          0.98        1.00        0.99         972

accuracy              0.99              0.99              0.99         1944
macro avg              0.99              0.99              0.99         1944
weighted avg              0.99              0.99              0.99         1944

SVM
=====
              precision    recall  f1-score   support

No Stroke      0.91        0.75        0.82         972
Stroke          0.79        0.92        0.85         972

accuracy              0.85              0.84              0.84         1944
macro avg              0.85              0.84              0.84         1944
weighted avg              0.85              0.84              0.84         1944

=====
STEP 6: MODEL COMPARISON TABLE
```

+ Code + Text

Variables Terminal

11:23 PM Python

Figure No: 04

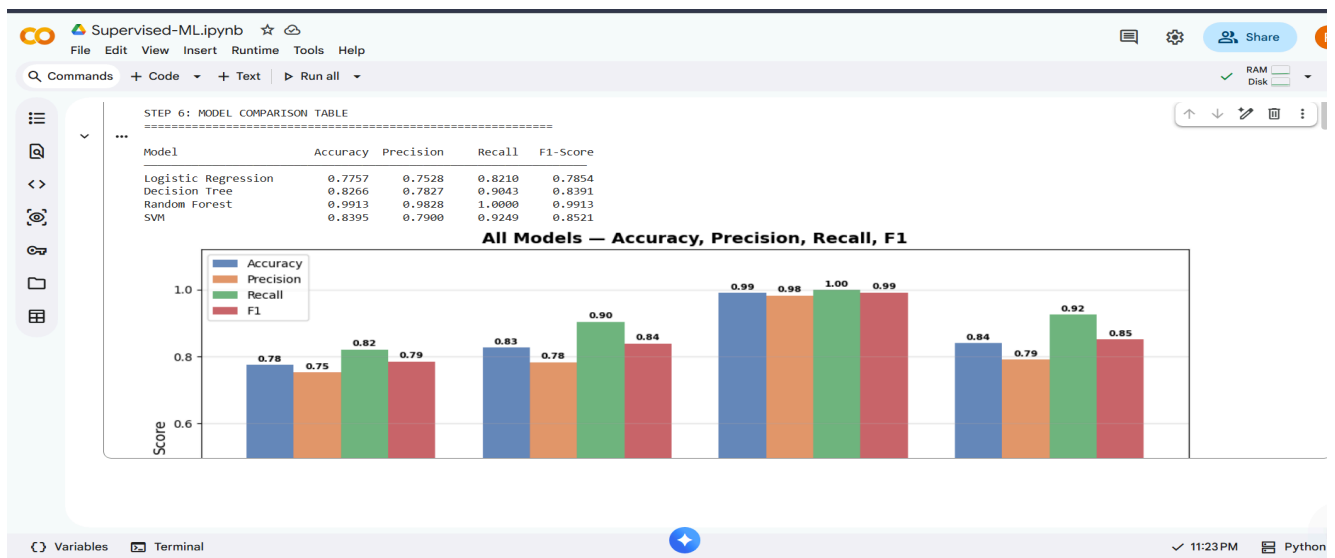


Figure No: 05



Figure No: 06

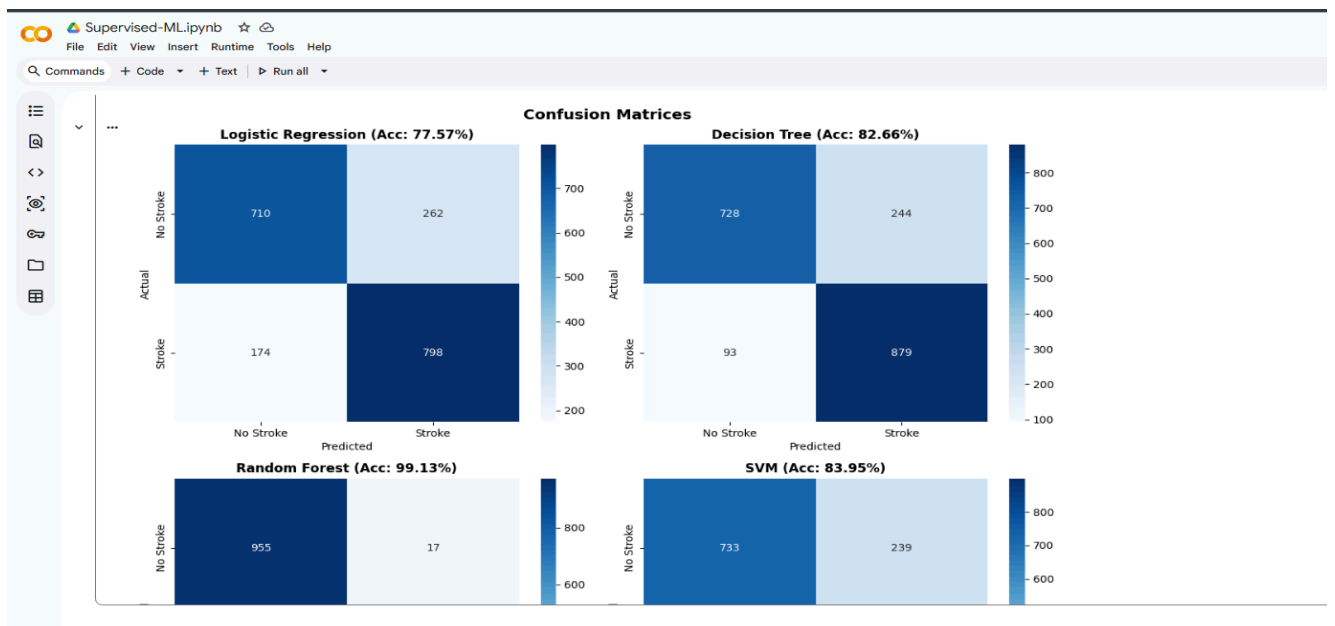


Figure No: 07



Figure No: 08



Figure No: 09

Conclusion:

This project successfully applied **four supervised machine learning algorithms** to predict stroke using a healthcare dataset. **After preprocessing and balancing the data**, all models were evaluated using standard performance metrics. Among them, **Random Forest** achieved the best results with the **highest accuracy and F1-score**. Overall, the project demonstrates that proper data preprocessing and model selection play a crucial role in building accurate machine learning prediction systems.